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synthesis and release stated earlier. This transmitter substance is synthesized in the presynaptic terminal from acetyl coenzyme A and choline in the presence of the enzyme *choline acetyltransferase*. Then it is transported into its specific vesicles. When the vesicles later release the acetylcholine into the synaptic cleft during synaptic neuronal signal transmission, the acetylcholine is rapidly split again to acetate and choline by the enzyme *cholinesterase*, which is present in the proteoglycan reticulum that fills the space of the synaptic cleft. And then again, inside the presynaptic terminal, the vesicles are recycled; choline is actively transported back into the terminal to be used again for synthesis of new acetylcholine.

1. Somatosensory function

The somatosensory system is a part of the sensory nervous system. The somatosensory system is a complex system of sensory neurons and neural pathways that responds to changes at the surface or inside the body.

The axons (as afferent nerve fibers) of sensory neurons connect with, or respond to, various receptor cells. These sensory receptor cells are activated by different stimuli such as heat and nociception, giving a functional name to the responding sensory neuron, such as a thermoreceptor which carries information about temperature changes.

Other types include mechanoreceptors, chemoreceptors, and nociceptors which send signals along a sensory nerve to the spinal cord where they may be processed by other sensory neurons and then relayed to the brain for further processing. Sensory receptors are found all over the body including the skin, epithelial tissues, muscles, bones and joints, internal organs, and the cardiovascular system.

SENSATION:

- *Sensation* is a conscious or unconscious awareness of external or internal stimuli:

Chemoreceptors, thermoreceptors, nociceptors, baroreceptors

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What are we not aware of?

X-rays, ultra high frequency sound waves, UV

light We have no sensory receptors for those

stimuli

- Perception is the conscious awareness & interpretation of a sensation. precisely localization & identification
- memories of our perceptions are stored in the cortex

Sensory Receptors:

Sensory receptors are specialized structures at the end of sensory neurons.

- Detectors for energy changes in the environment.
- Transducers as they transform energy changes to action potentials.

Classification of Receptors:

According to specific type of energy needed to stimulate them:

1- Mechanoreceptors:

- . Touch receptors.
- . Pressure receptors.
- . Proprioceptors.

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- . Vibration receptors.

- . Vestibular receptors.

2- Thermoreceptors.

3- Photoreceptors: Rods and cones in retina.

4- Nociceptors: Free nerve ending in skin & mm

5- Chemoreceptors:

- . Chemosensitive cells in olfactory mm.

- . Taste buds.

- . Osmoreceptors in hypothalamus.

- . Glucoreceptors in hypothalamus.

Classification by Location:

1. Exteroceptors

near surface of body

receive external

stimuli

hearing, vision, smell, taste, touch, pressure, pain, vibration & temperature

2. Interoceptors

monitors internal environment (BV or viscera)

not conscious except for pain or pressure

3. Proprioceptors

muscle, tendon, joint & internal ear

senses body position & movement

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Structural Classification of Receptors (table 3)

1. <i>Unencapsulated</i>	2. <i>Encapsulated</i>	3. <i>Sensory cells</i>
free nerve endings	Meissner's corpuscles	vision
Merkel discs	Krause's end bulbs	taste
root hair plexus	Pacinian corpuscles	hearing
	Ruffini's corpuscles	Balance
	Muscle spindles	
	Golgi tendon organs	

Mechanism of Stimulation of Receptors:

1- Receptor potential.

2- Spread of receptor potential.

3- Action potential.

4- Frequency of action potential is directly proportional to logarithm intensity of the stimulus.

Adaptation of Sensory Receptors:

Change in sensitivity to long-lasting stimuli

decrease in responsiveness of a receptor

bad smells disappear

very hot water starts to feel only warm

potential amplitudes decrease during a maintained, constant stimulus

Variability in tendency to adapt:

Rapidly adapting receptors (smell, pressure, touch)

specialized for detecting changes

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Slowly adapting receptors (pain, body position)

nerve impulses continue as long as the stimulus persists – Pain is not easily ignored.

Ascending Tracts (Pathway of Sensations) (figure 12):

Pathway of Crude Touch

1. Receptors: Free nerve ending & hair end organ
2. Transmission: A delta fibres.
3. 1st order neuron.
4. 2nd order neuron: Ventral spinothalamic tract.
5. 3rd order neuron: From thalamus to sensory area of cerebral cortex.

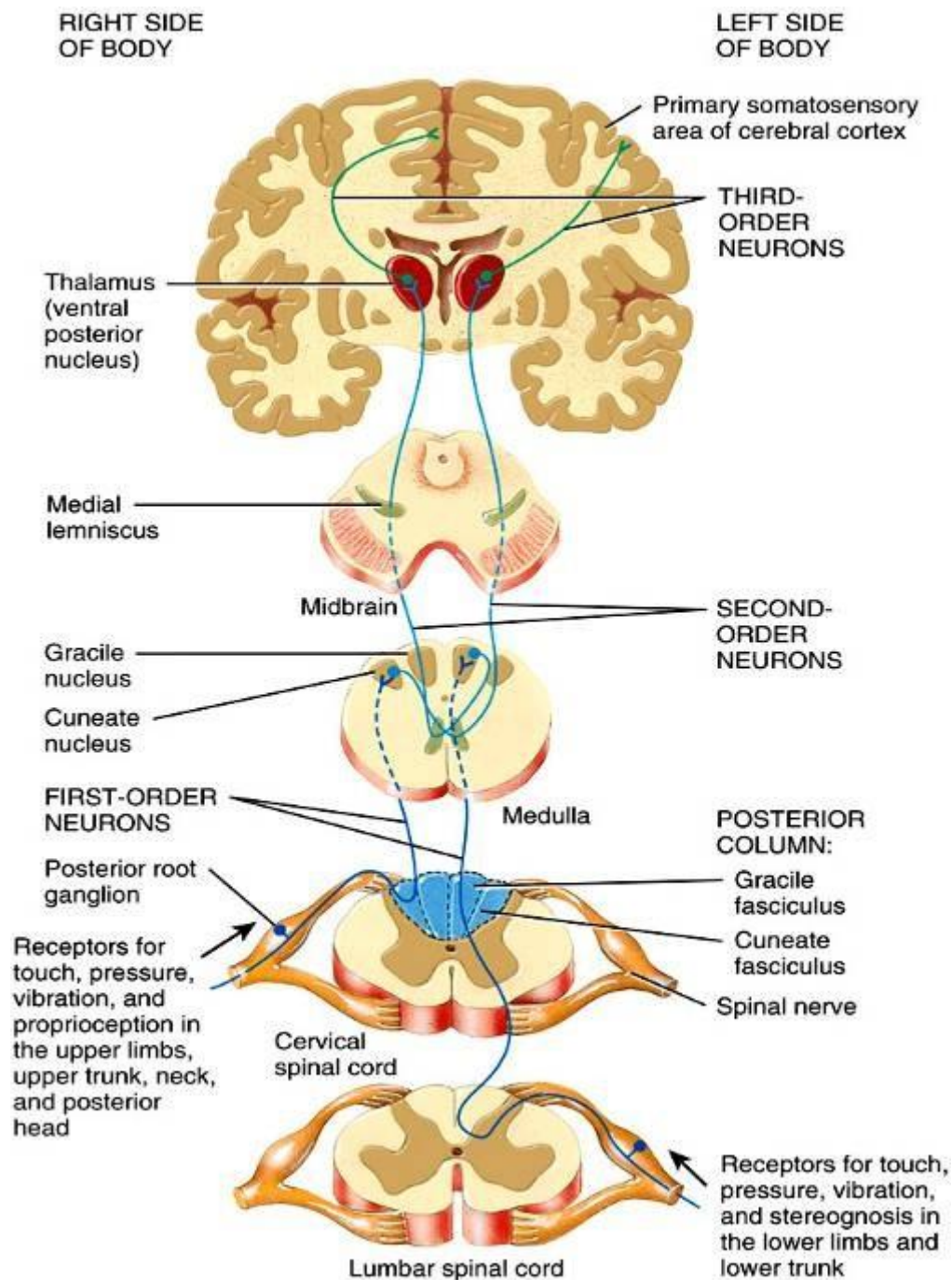


Figure 12 : tactile pathway

Pathway of Pain and Thermal sensations (figure 13):

- 1st order neuron: To posterior horn cells of spinal cord.
- 2nd order neuron: Along lateral spinothalamic tract of opposite side.

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- 3rd order neuron: Arise from thalamus to sensory area of cerebral cortex.

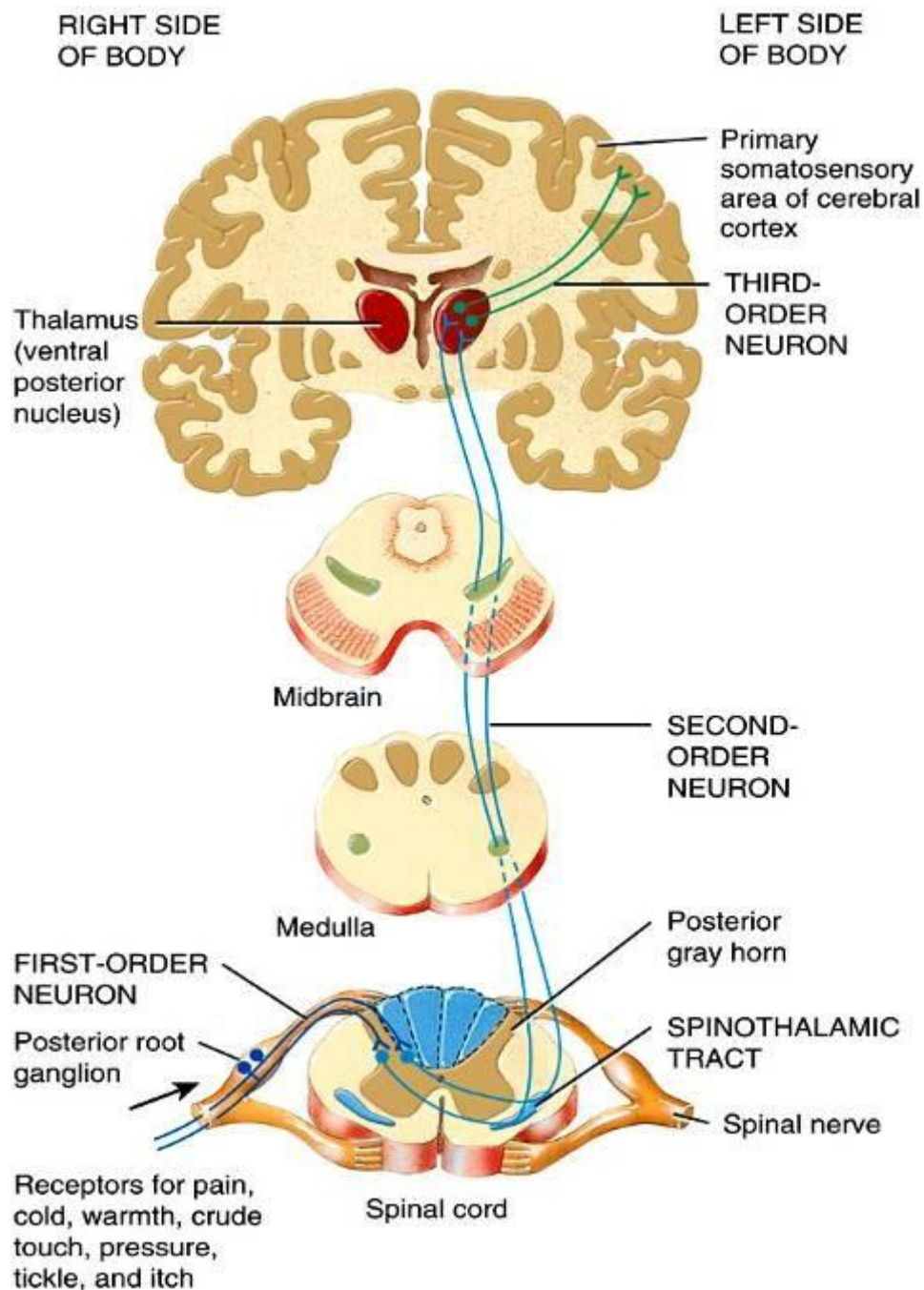


Figure 13: pathway of pain and thermal sensations

Sensations From The Face And Head:

The Trigeminal nerve is formed of three branches (figure 14):

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1. The Ophthalmic branch: carries sensory impulses from the forehead, cornea, conjunctiva, and mucosa of the nose and air sinuses.
2. The Maxillary branch: carries sensory impulses from the nasal septum, hard palate, upper teeth and gums as well as from the skin of the cheeks.
3. The Mandibular branch: carries somatic sensory impulses from the lower teeth and gums, anterior 2/3 rd of the tongue, the inside of the cheek and lower lip, as well as the skin over the mandible.

The Trigeminal nerve also carries pain sensory impulses from the intracranial pain sensitive structures above the Tentorium cerebelli.

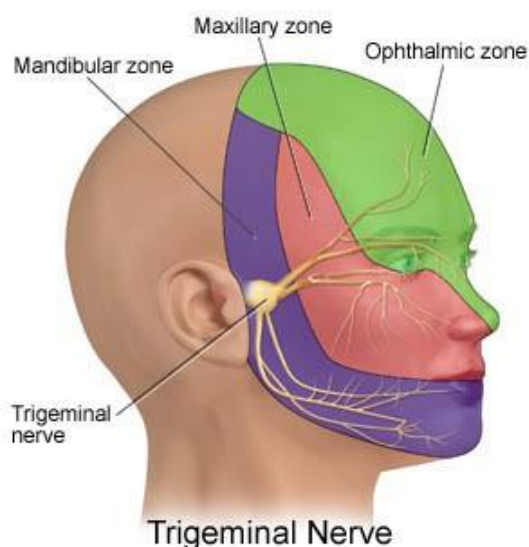


Figure 14: trigeminal nerve

Sensory pathway from the face (figure 15)

- The first order neuron: -
- The afferent fibers of the Trigeminal nerve carrying somatic sensory impulses from the head and face are the peripheral branches of the

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unipolar cell bodies of the Gasserian ganglion of the Trigeminal nerve which corresponds to the DRG of spinal nerves.

- The central branches of these cells form the sensory root of Trigeminal nerve which enter the CNS at the pons and divide into two branches; an ascending and a descending branch.
- The ascending branch ends on the Main Sensory Nucleus of Trigeminal in pons. The fibers carry sensory impulses for touch, vibration, and kinesthesia. Therefore, re Main Sensory Nucleus of Trigeminal corresponds to the Gracile and Cuneate nuclei in the medulla.
- The descending branch carries the sensory impulses for pain and temperature and forms the bulbospinal tract that ends on the Spinal Nucleus of Trigeminal present in pons, medulla and extending down to the second cervical segment of spinal cord. The Spinal Nucleus of Trigeminal corresponds to the Substantia Gelatinosa of Rolandi (SGR).

The second order neuron:

- Axons of the MSN and SN of Trigeminal nerve form the 2nd order neuron that crosses the midline and ascend in the opposite side of the brain stem forming the Trigeminal Lemniscus that ends on the PMVNT.
- There is evidence that not all fibers cross the midline and some of them ascend in the same side forming Ipsilateral representation.
- Clinical Significance: - In hemiplegia, the loss of sensation from the face is usually incomplete.

The third order neuron:

- Axons of PMVNT pass in the posterior limb of the internal capsule → sensory radiation → sensory area of cerebral cortex .

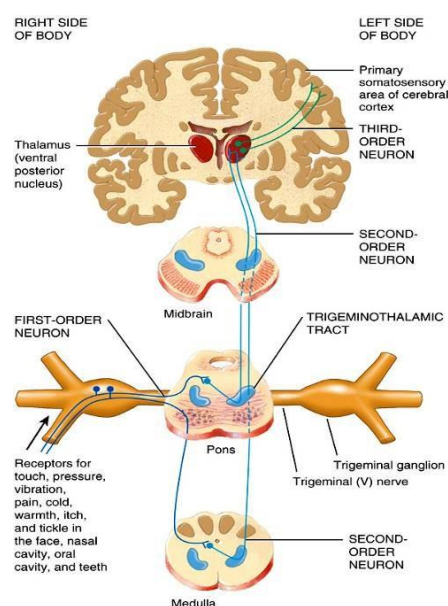
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- Clinical Significance:

Trigeminal Neuralgia: Attacks of severe pain of unknown cause (irritation) referred to the area of distribution of the trigeminal nerve.

Trigeminal Anaesthesia: There is loss of sensation in the area supplied by the Trigeminal nerve due to a lesion in the Gasserian ganglion. Corneal anaesthesia is dangerous as it leads to corneal ulcer.

Figure 15 sensory pathway from the face



HEADACHE

- **Definition:** It is a type of referred pain to the surface of the head from deep structures.
- **Causes:** It may be due to intracranial or extracranial causes.

1. Intracranial headache: Intracranial pain sensitive structures: -

- A. The cerebral and dural arteries specially the middle meningeal artery.
- B. The dura at the base of the brain.
- C. The tentorium.
- D. The great venous sinuses.

Areas of the head to which intracranial headache is referred

- Impulses from pain sensitive structures above the tentorium are

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transmitted along the Trigeminal nerve and pain is referred to the skin areas supplied by the

Trigeminal nerve i.e. the upper part of the head in front of the ear → Frontal headache.

- Impulses from pain sensitive structures below the tentorium are transmitted along the posterior roots of the 2nd cervical spinal nerve and pain is referred to the skin area supplied by the 2nd cervical posterior root i.e. the posterior part of the head Occipital headache.

Causes of intracranial headache

- Meningeal irritation: Due to inflammation (meningitis) or trauma.
- Brain tumours: Due to traction and irritation of meninges by over stretch.
 1. Tumors above the tentorium → Frontal headache.
 2. Tumours below the tentorium → Occipital headache.
- Decreased cerebrospinal fluid (CSF) pressure: Removal of as much as 20 ml of CSF → sinking of the brain -> traction on meninges specially in the erect posture → headache. It is very common following spinal anaesthesia.
- Hypertension: Due to excessive stretch of the walls of intracranial arteries with each cardiac cycle. It is related to the magnitude of pulse pressure.
- Migraine headache: The cause is unknown but stress and emotional disturbances contribute and the sequence of events is as follows:
 1. Sudden attack of vasospasm of cerebral arteries → ischemia of certain areas of the brain → prodroma (aura) of blurring of vision,

and hallucinations about half an hour before headache starts.

2. Ischemia sensory + low O₂ tension vasodilatation → excessive distension and stretch of cerebral arteries → severe headache.

- Alcoholic headache: Alcohol intoxication produces headache through: meningeal irritation and cerebral vasodilatation.
- Constipation: Leads to absorption of toxic products of fermented and putrefied food materials in the large intestine → meningeal irritation.

2. Physiology of Pain (I)

For scientific and clinical purposes, pain is defined by the International Association for the Study of Pain (IASP) as, “an unpleasant sensory and emotional experience associated with actual or potential tissue damage, or described in terms of such damage.” This is to be distinguished from the term nociception which the IASP defines as the unconscious activity induced by a harmful stimulus applied to sense receptors. Pain is very important warning sensation that is produced whenever any tissue is damaged.

It initiates protective reflexes for removing the injurious stimulus.

Pain receptors:

- They are free nerve endings (FNE).
- They are widely

distributed: In the skin →

cutaneous pain.

In deep tissues → deep pain.

In viscera → visceral pain.